

Sr Automation Engg/Sr SDET Role Requirement

Senior SDET (Software Development Engineer in Test) - Role Requirements

1. Role Overview

The **Senior SDET** is a key player in ensuring software quality, responsible for **test automation, performance testing, and continuous integration** within the Agile team. The role requires a deep understanding of **test automation frameworks, software development best practices, and DevOps processes**.

This position involves:

- Designing and implementing **scalable, maintainable, and high-performing** automated testing frameworks.
- Ensuring **end-to-end testing coverage** for functional, integration, regression, and performance testing.
- Collaborating with developers, product owners, and DevOps teams to **streamline automation pipelines**.

2. Key Responsibilities

- **Test Strategy & Planning**
 - Define and implement comprehensive **test automation strategies** for web, API, database, and performance testing.
 - Identify **test scenarios, acceptance criteria, and performance benchmarks** based on functional and non-functional requirements.
 - Design test strategies for **Agile development cycles**, ensuring integration with DevOps CI/CD pipelines.
- **Automation Development**
 - Develop and maintain **scalable and maintainable** test automation frameworks using **Selenium, Cucumber**.
 - Write high-quality, reusable **test scripts** in **Java**, following Object-Oriented programming principles.
 - Implement and execute **test automation** for functional, regression, integration, and performance testing.
- **Performance & API Testing**
 - Work with performance engineers to optimize software quality using tools like **BlazeMeter, JMeter, Taurus**.
 - Develop and maintain API automation test scripts for **REST and SOAP services** using **Postman and REST Assured**.
- **Continuous Integration & DevOps**
 - Integrate test automation into the **CI/CD pipeline** using **Jenkins, Azure DevOps, or TeamCity**.
 - Ensure automated tests are triggered in the pipeline and provide detailed test results with failure analysis.
 - Work with the **DevOps team** to optimize test execution speed and stability in CI/CD environments.
- **Bug Tracking & Resolution**
 - Identify, isolate, document, regress, and track defects using tools like **JIRA & QTest**.
 - Conduct **root cause analysis (RCA)** and collaborate with developers to resolve issues.
- **Agile & Team Collaboration**
 - Participate in **Agile ceremonies** (Sprint Planning, Daily Standups, Retrospectives) as a key testing stakeholder.
 - Advocate **test-driven development (TDD), behavior-driven development (BDD)**, and risk-based testing practices.
 - Assist in **manual testing efforts** when needed, focusing on exploratory testing.

3. Required Skills & Expertise

A. Automation Testing & Development

- **Programming Languages:**
 - Proficient in **Java** or **C#** (Optional)
 - (Object-Oriented Programming).
 - Hands-on experience with **Python, JavaScript, or PowerShell** for scripting.
- **Automation Frameworks:**
 - Strong experience with **Selenium, Cucumber**.
 - Familiarity with **NUnit, TestNG, or xUnit** for test case execution.
- **API Testing:**
 - Proficiency with **Postman, REST Assured, or SoapUI**(Optional).
 - Ability to automate API testing and validate request-response cycles.
- **Performance Testing:**
 - Experience with **BlazeMeter, JMeter, Taurus** to identify system bottlenecks.
 - Ability to analyze performance results and work with developers to resolve issues.

B. Database & Batch Testing

- Strong **SQL skills** for **database validation and batch processing testing**.
- Experience working with **Oracle or Sybase, SQL Server (Optional)**.
- Ability to write **complex SQL queries** to validate back-end data integrity.

C. CI/CD & DevOps Integration

- Hands-on experience with **Jenkins, Azure DevOps(Optional)**.
- Ability to integrate automation suites into **CI/CD pipelines**.
- Exposure to **Docker, Kubernetes**, and cloud-based test environments.

D. Agile & Collaboration

- Strong understanding of **Agile Testing Quadrants and Risk-Based Testing**.
- Knowledge of **TDD, BDD, ATDD** practices.
- Experience in Agile/Scrum environments with **cross-functional teams**.

4. Preferred Skills (Nice to Have)

- Experience with **mobile automation** testing (Appium or equivalent).
 - Exposure to **Cloud-Based Testing** (AWS, Azure, GCP).
 - Basic knowledge of **Unix/Linux** environments.
 - Financial services or CRM domain experience.
 - Knowledge of **monitoring tools** like Splunk or New Relic.
- ### 6. Role Expectations
- **Leadership & Ownership:**
 - Own and drive **test automation strategy and implementation** for the team.
 - Mentor junior SDETs and **ensure best practices** in automation.
 - Work proactively to **identify gaps** in the test coverage and improve automation efforts.
 - **Strategic Thinking & Problem-Solving:**
 - Identify potential testing bottlenecks early and **work on solutions**.
 - Innovate by introducing **new tools and technologies** for efficiency improvements.
 - Lead initiatives for **automation adoption and shift-left testing**.
 - **Quality Advocacy & Agile Mindset:**
 - Foster a **quality-first approach** by implementing **continuous testing**.
 - Champion **Agile, BDD, and TDD practices** to optimize test development.
 - Drive **collaboration with developers and product owners** to improve testability.